Foundations of the human body

Practice Paper 1 | 30 marks | Suggested time: 35 minutes

Teacher/student marking guidance. Award credit for equivalent scientifically correct wording. Marks within each part are capped at the stated total.

Question 1 Aerobic respiration — 7 marks

a) State the word equation for aerobic respiration. [2]

- Glucose + oxygen on the left (1).
- Carbon dioxide + water on the right (1).

b) Give TWO different uses of the energy released by aerobic respiration. [2]

- Any two: muscle contraction, growth, repair, maintaining temperature, active transport or other valid life process (1 each).

c) Explain how glucose from a meal and oxygen from the lungs reach a muscle cell. [3]

- Carbohydrates are digested into glucose and glucose is absorbed through the small intestine into blood (1).
- Oxygen diffuses from alveoli into blood (1).
- The heart pumps blood carrying both substances to capillaries near muscle cells (1).

Question 2 Digestion and nutrients — 8 marks

a) State ONE function of each of these organs: mouth, oesophagus, stomach, small intestine. [4]

- Mouth: chewing physically breaks food into smaller pieces or saliva begins chemical digestion (1).
- Oesophagus: peristalsis moves food to the stomach (1).
- Stomach: churns food or produces acid and enzymes for protein digestion (1).
- Small intestine: digestion/absorption of nutrients (1).

b) State ONE main function of each nutrient: protein, lipid, fibre. [3]

- Protein: growth and repair or builds body tissues (1).
- Lipid: energy store or cell membrane component (1).
- Fibre: helps material move through the digestive tract (1).

c) Why must many large food molecules be digested before they can be absorbed? [1]

- They must be broken into smaller, soluble molecules that can cross the gut wall (1).

Question 3 Digestion and absorption — 8 marks

a) Explain the difference between physical and chemical digestion. [2]

- Physical digestion changes the size of food pieces without chemically breaking molecules (1).
- Chemical digestion uses enzymes to break large food molecules into smaller molecules (1).

b) Explain TWO ways the structure of villi makes nutrient absorption efficient. [4]

- Large total surface area (1) allows more molecules to pass through per unit time (1).
- Thin wall (1) provides a short diffusion distance (1).
- Allow alternative correct linked pair: many capillaries (1) carry absorbed substances away and maintain a concentration gradient (1). Max 4.

c) Define diffusion, including the direction of net particle movement. [2]

- Net movement of particles (1) from a region of higher to lower concentration/down a concentration gradient (1).

Question 4 Gas exchange — 7 marks

a) Name the tiny air sacs where most gas exchange occurs in the lungs. [1]

- Alveoli (1).

b) State the direction in which oxygen and carbon dioxide diffuse between alveoli and capillary blood. [2]

- Oxygen: from alveoli into blood (1).
- Carbon dioxide: from blood into alveoli (1).

c) Explain THREE features that help alveoli exchange gases efficiently. [3]

- Large surface area to allow more diffusion (1).
- Thin walls giving a short diffusion distance (1).
- Rich capillary supply/maintained concentration gradient (1).

d) Why is oxygen entering the blood important for cells? [1]

- It is a reactant needed for aerobic respiration to release energy (1).

Total: 30 marks. Mark each valid point once unless stated otherwise.